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DOCUMENT 4 — HEAT PUMP

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Component: Heat Pump

Purpose:

Upgrade low-temperature heat to district heating supply temperature.

Inputs:

- Electrical input power P_{el} [W]
- Source temperature $T0_K$ [K]

Outputs:

- Useful heat flow Q_{heat} [W]
- Supply temperature T_{supply_K} [K]

Key Variables:

COP : coefficient of performance [-]

$T0_K$: source temperature [K]

T_{supply_K} : supply temperature [K]

Equations:

$$Q_{heat} = COP * P_{el}$$

$$COP = T_{supply_K} / (T_{supply_K} - T0_K)$$

$$COP = COP$$

$$T0_K = T0_K$$

$$T_{supply_K} = T_{supply_K}$$

Assumptions:

- Idealized steady operation
- Carnot-based approximation